

Homework 20

I posted this draft homework instead of the final version by mistake! I'm sorry that it repeated a problem that I moved to an earlier assignment.

1. Please read Section 6.3 of Durrett.
2. Twelve ducks fly overhead. Each of 6 hunters picks one duck at random to aim at and kills it with probability 0.6. What is the expected number of ducks that are killed? When we met, we started to answer this question in one way. We were calculating the probability that, say, duck 1 is killed. We conditioned on the number of hunters who aimed at duck 1 (two-stage experiment), which led to a big calculation with binomial probabilities. Here is a method that you'll like better.
I repeated this problem by mistake. See solutions for Homework 19.
 - a. What is the probability that hunter 1 hits duck 1? What is the probability that hunter 1 does not hit duck 1?
 - b. What is the probability that none of the hunters hits duck 1? What is the probability that at least one of the hunters hits duck 1? This second probability is the probability that duck 1 is killed.
 - c. What is the expected number of ducks that are killed? Explain using the method of indicators.
3. (Review) Imagine you shuffle a standard deck of cards and start drawing from the shuffled deck. After every draw, you return the card to the deck and reshuffle. Let T be the random variable representing the number of draws until you get your first ace.
 - a. What distribution does T have? (It is one of our standard distributions from earlier in the semester.)
 $T \sim \text{geometric}(4/52)$
 - b. What is the expected number of draws until you get your first ace, i.e. what is $E[T]$?
 $E[T] = 1 / p$, where p is $4 / 52$. That is $52 / 4 = 13$ draws.
4. Imagine you shuffle a standard deck of cards.
 - a. What is the probability that the queen of spades comes before all four aces in the shuffled deck? (This probability, of course, would be the same for any card that is not an ace.)

Durrett would tell us to calculate the probability in the following way:

Only the position of the five named cards (queen of spades and the four aces) matters. The queen of spades could be any of five positions relative to the other cards: first, second, third, fourth, or fifth. It is equally likely to be in any of these positions, and so the probability is $1/5$.

- b. Suppose we draw cards from the deck and stop when we first see an ace. Let N be the random variable representing the number of cards we draw before that first ace. Explain how to write N as a sum of 48 identically distributed (but not independent) random variables.

Number the 48 non-ace cards as 1, 2, 3, ..., 48. Let A_i be the event that non-ace card i comes before all the aces. Just as with the queen of spades in part a, we have $P(A_i) = 1/5$. Let X_i be the indicator for the event A_i . Then $X_i \sim \text{Bernoulli}(1/5)$.

We can write N as $X_1 + X_2 + \dots + X_{48}$: For any outcome (arrangement of the 52 cards), the number X_i random variables with value 1 is exactly the number of events A_i that occur, i.e. the number of non-ace cards before all the aces. Counting the number of 1s among the X_i values is the same as summing $X_1 + \dots + X_{48}$.

- c. Calculate $E[N]$, the expected number of cards we draw before our first ace.

We have $E[N] = E[X_1] + E[X_2] + \dots + E[X_{48}]$ by the additivity of expected value. For each i , we have $E[X_i] = P(A_i) = 1/5$, and so $E[N] = 48(1/5) = 48/5 = 9.6$ cards.

The expected number of cards before the first ace is 9.6. We can compare this value to what we calculated in question 3. The expected number of cards to get the first ace is 13, if we replace cards after drawing. The expected number of cards before the first ace is thus 12, if we replace cards after drawing.

5. A team plays a 162-game regular season. We view the team's number of wins X as a random variable. As a first model for X , assume that the outcomes of the games are independent and that the team has a 0.5 probability of winning any game it plays.

- a. Express X as a sum of 162 independent and identically distributed (iid) random variables. What is the common distribution of these 162 variables? Calculate its expected value and variance.

Let B_i be the event that the team wins game i . Let Y_i be the indicator for the event A_i . We then have $X = Y_1 + \dots + Y_{162}$.

Each random variable Y_i has $Y_i \sim \text{Bernoulli}(1/2)$. Therefore, $E[Y_i] = 1/2$ and $\text{Var}(Y_i) = 1/4$.

- b. Use your formula for X as a sum of 162 iid random variables to calculate $E[X]$, $\text{Var}(X)$, and $\sigma(X)$ (standard deviation of X).

By the additivity of expected value, we have

$$E[X] = E[Y_1] + \dots + E[Y_{162}] = 162(1/2) = 81.$$

Since the events A_i are independent, so are the random variables Y_i . Therefore,

$$\text{Var}(X) = \text{Var}(Y_1) + \dots + \text{Var}(Y_{162}) = 162(1/4) = 40.5.$$

$$\sigma(X) \approx 6.36.$$

We expect the team to win around 81 games give or take 6 games.

6. Suppose we roll a die 10 times and sum the numbers that we roll. Let Y be the random variable representing the sum.

a. Express Y as a sum of 10 iid random variables. What is the common distribution of these 10 variables? Calculate its expected value and variance.

We can write $Y = Z_1 + \dots + Z_{10}$, where Z_i is the number from roll i . The random variables Z_i take values $\{1, 2, 3, 4, 5, 6\}$, each with probability $1/6$. Durrett calculates the expected value and variance for this distribution in Chapter 1. We can look the numbers up there (or recalculate):

$$E[Z_i] = 7/2 = 3.5$$

$$\text{Var}(Z_i) = 105/36 \approx 2.9167$$

b. Use your formula for Y as a sum of 10 iid random variables to calculate $E[Y]$, $\text{Var}(Y)$, and $\sigma(Y)$.

By the additivity of expected value, $E[Y] = E[Z_1] + \dots + E[Z_{10}] = 10(3.5) = 35$.

Since the random variables representing the different rolls are independent, $\text{Var}(Y) = \text{Var}(Z_1) + \dots + \text{Var}(Z_{10}) = 10(105/36) = 525/18 \approx 29.167$

$$\sigma(Y) = \sqrt{\frac{525}{18}} = \frac{5}{3} \sqrt{\frac{21}{2}} \approx 5.401$$

We expect the sum of the ten rolls to be 35 give or take 5.